Nutthawat Panyangnoi

Title: *Firmware Wizard Application – Informal Questions*

Q1. Show us a cool thing you've built/coded up

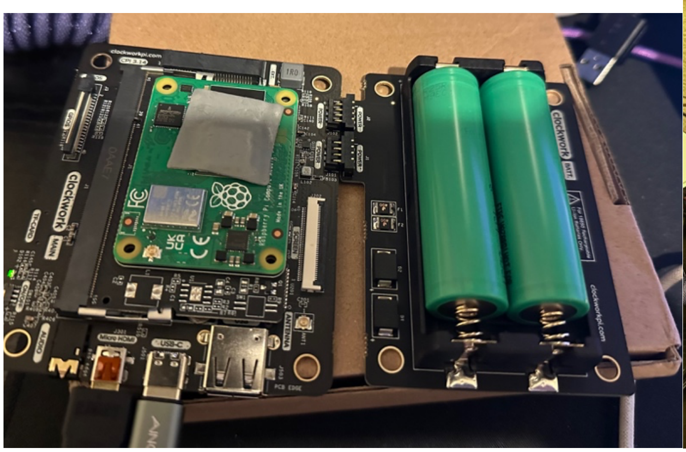
First project that I want to present is me customize Linux distro built with the Yocto Project tool. My OS is named Mirai Linux in Japanese it's mean "Future" which is based on Debian. It is currently only support on ARM64 bits on Raspberry pi CM4 and Raspberry Pi 4. My OS is made with the love of Linux. Although, I ran into many problems while creating an OS. I am planning to put this OS into a handheld PC which is my senior project that I worked with two classmates. My current operating system version is still barebone, it needs a lot of improvement such as installation package, desktop environment like xfce, gnome, etc.

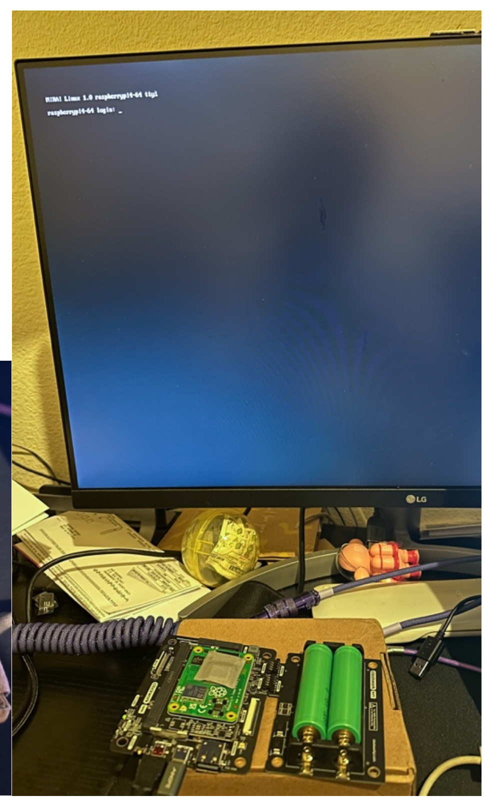
At first, I was planning to create an OS from scratch but because of I lacked knowledge in writing kernel and many aspects in OS design. I come across multiple issue, so I did give up and build with alternative tool like Yocto Project for now.

My first attempt: https://github.com/kur0neko/MiraiLinux

You can download OS image RPI-SDIMG here and run it on QEMU (ARM64 bits)

My senior project handheld PC with help of Clockwork Pi Board.



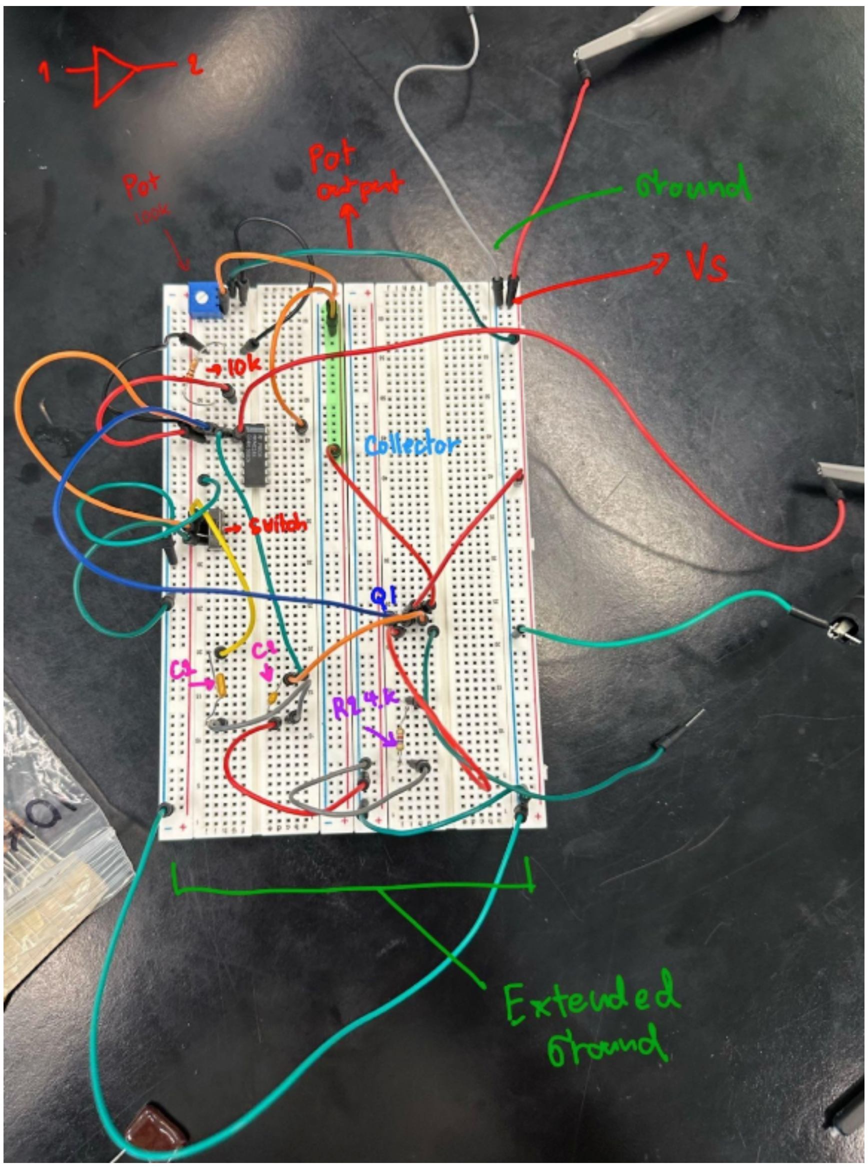


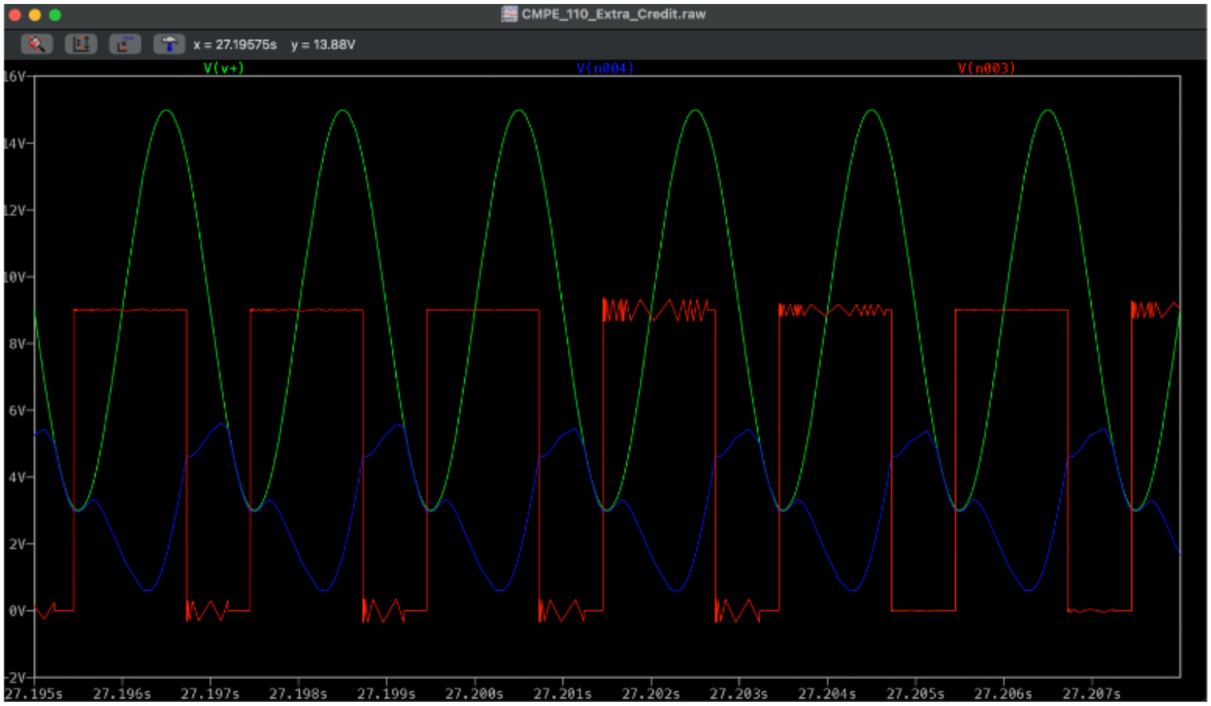
Second project that I want to present is an engineer lab class project with my classmate, we built Square and Triangle wave generator using Schmitt Trigger Op-amp.

My role is designing the circuit on LTspice and construct the breadboard.

[Video Link]

[GitHub Link]



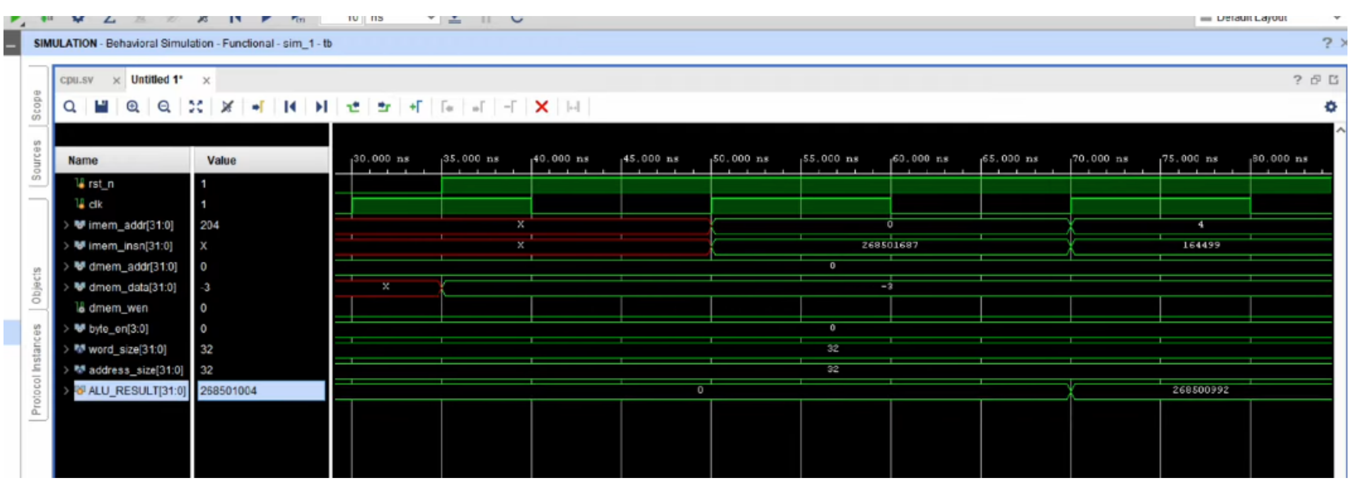


Simulate waveform

Third Project is also an engineer lab class project that I collaborated with classmate. We implemented FPGA pipeline design RISC-V Architecture using AMD Vivado and Verilog language.

We split into multiple module for each members, then we put every module together to create the simulate RISC-V pipeline

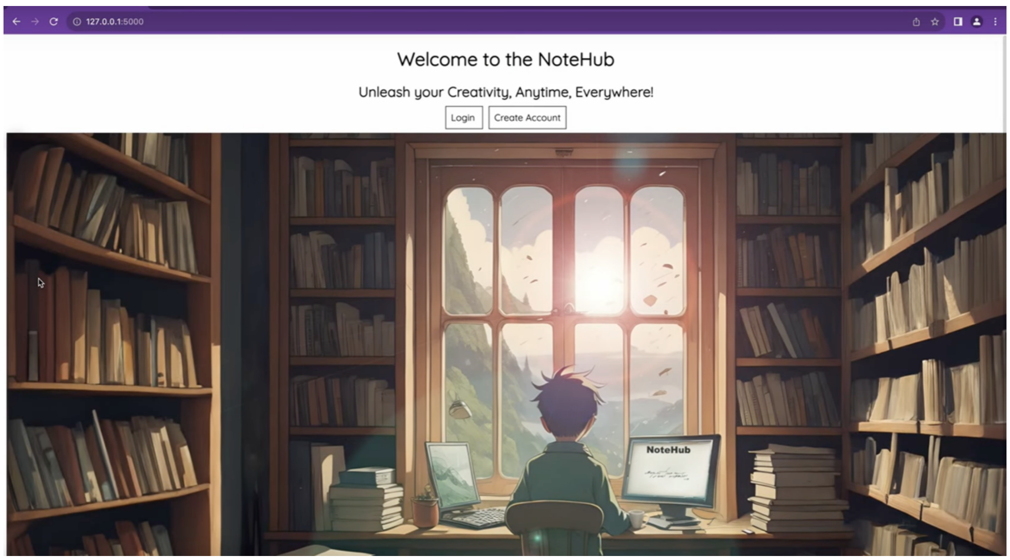
Document and code can be finding here [GitHub Link] and [Video Link]



Fourth Project is also a class project that I worked with classmate. We implemented Full Stack website based on Python3 Frameworks like Django, Flash, SQL alchemy and Jinja Web template engine. Our project is inspired by Notion application.

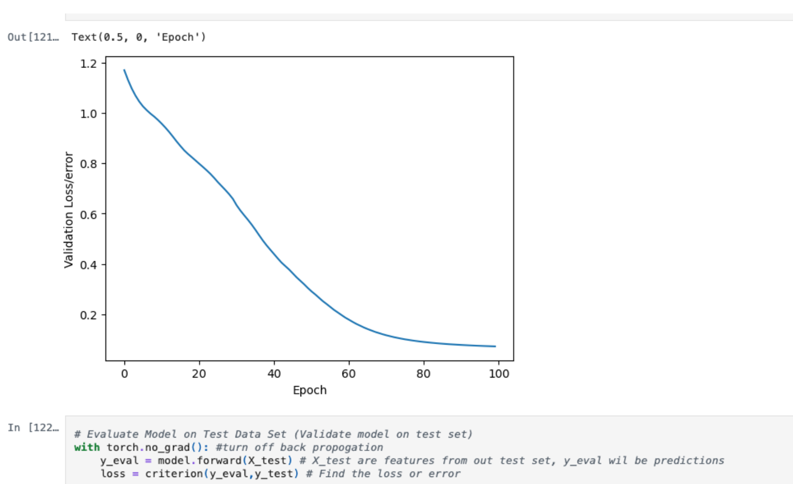
GitHub Link

Video Link



The last project is my personal project when I self-learning machine learning, Deep learning and AI. I designed and implemented a Neural Network Project using PyTorch, Numpy, Scikit-Learn, to classify iris flowers into Setosa, Versicolor, or Virginica species based on sepal and petal measurement. Credit date set from https://gist.github.com/netj/8836201

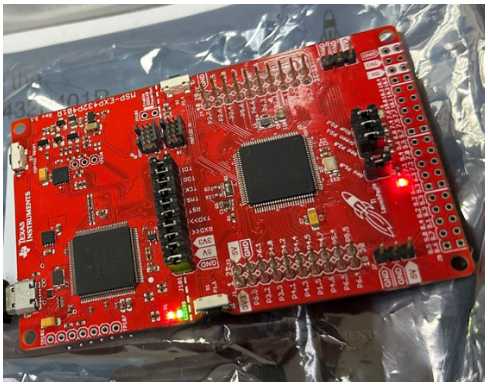
GitHub Link



Q2. Favorite niche peripheral/feature of a microcontroller?

There were two cool MCU that I have experiences to work with

1. TI MSP432 P401R



During embedded class I got to write C programs toggles with this board. Some features that I like the most about this board were this board have ARM Cortex architecture allow it to perform in many powers' mode such as low power mode (LPM0-LPM4.5) and can switch to high power mode as well. This board have multiple oscillators internal and external allow it to choose between low frequencies and higher frequencies, the ADC of this board can sample up to 14-bit resolution and this board also come with multiple system timers like Timer_A and Timer32. Another interesting feature this board can support up to 84 GPIO pins where 16 GPIO pins are on high power.

2. Astraeus-1 Board (Customed board made for satellite equipped with XBee Pro S38 radio, barometer Mel3115, motion tracker ICM 20948, GPS, NEO M9N rope meter level GNSS position and Micro Mode MCU)

   This is the custom board that I get to learned during my time at Cube3 club where we built satellite and sent to space. I join the club very late; I only got a chance to build some small feature for this project like output data into graph on web application home page using Next.js. I never get to see the hardware. However, the spec sheet data that I had is this board called Astraeus-1. With all chips on this MCU made it super interesting. It consists with multiple sensors like barometer, motion tracker, GPS, thermal sensor, etc. I found it very cool get to see a custom robust MCU.

Q3. Sketchy firmware "fix" you pulled off

While working on TI MSP432 P401R, I came across many situations that I accidentally accessing invalid memory address or access improper member address. The MMU was supposed to block this from happening but occasionally it still let my code pass through restricted memory regions. My most sketchy way was using the watchdog timer. If the CPU stalled or jump into invalid segment of loop watchdog will force a reset and reload the last safe state. But sometimes I also add catch-add hard reset function, if the system crashed, this would force to reset and reload state before brick.

Q4. Debugging a PCB that resets randomly

If I follow step of the board manual to do a hard reset, factory reset and if it did not work out. I would check if the firmware were latest up to date. Next, I will try to flash memory. Sometimes the code can corrupt the memory region. If all the software update, and reset doesn't work. I will start to perform hardware diagnostic. I would check each component with DMM or oscilloscope to verify the power draw and clock signal. Then, I will look for short circuit, overheating regulators or a component failure.

Q5. Which skill categories from the role apply to you most

Due to my background is heavily on software development and Linux. I think these are my strongest skillset. I've built a lightweight Linux for Arm architecture, FPGA design for RISC-V, know how to write assembly in MASM and NASM. Write low level C program to toggle with MCU and skill to diagnose electrical component problem. While I never have any experience in write a firmware before I do believe that with my background and experience will quickly help me catch on understand the fundamental of writing a firmware. With the opportunity to work with Windborne will allow me to improve my technical skill in writing the firmware and further understanding the fundamental of firmware better in the future and become a proper engineer.